LN8 Arbitrage Pricing

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Spring 2023

Outlines

Textbook: Arbitrage Theory in Continuous Time: Chapter 7

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Introduction

Definition 1

The price process B is the price of a **risk free** *asset if it has the dynamics*

$$dB(t) = r(t)B(t)dt, \tag{1}$$

where r is any adapted process.

Definition 2

The **Black–Scholes model** *consists of two assets with dynamics given by*

$$dB(t) = rB(t)dt, \tag{2}$$

$$dS(t) = \alpha S(t)dt + \sigma S(t)d\bar{W}(t), \tag{3}$$

where r, α and σ are deterministic constants.

European Call Option

Definition 3

A European call option with exercise price (or strike price) K and time of maturity (exercise date) T on the underlying asset S is a contract defined by the following clauses.

- *The holder of the option has, at time T, the right to buy one share of the underlying stock at the price K from the underwriter of the option.*
- *The holder of the option is in no way obliged to buy the underlying stock.*
- *The right to buy the underlying stock at the price K can only be exercised at the precise time T.*

Contingent Claims

- A **European put option** is an option which in the same way gives the holder the right to **sell** a share of the underlying asset at a predetermined strike price.
- For an **American call option** the right to buy a share of the underlying asset can be exercised at any time before the given time of maturity.
- All these contracts are called **derivative instruments** or **contingent claims**.

Formal Definition of A Contingent Claim

Definition 4

Consider a financial market with vector price process S. A **contingent claim** *with date of maturity (exercise date) T, also called a T-claim, is any stochastic variable $\mathcal{X} \in \mathcal{F}_T^S$. A contingent claim $\mathcal{X}$ is called a simple claim if it is of the form $\mathcal{X} = \Phi(S(T))$. The function Φ is called the* **contract function**.

Example European call option

$$\Phi(x) = max[x - K, 0].$$

Arbitrage

Definition 5

An **arbitrage** *possibility on a financial market is a self-financed portfolio h such that*

$$\begin{aligned} V^h(0) &= 0, \\ P(V^h(T) \geq 0) &= 1. \\ P(V^h(T) > 0) &> 0. \end{aligned} \tag{4}$$

We say that the market is **arbitrage free** *if there are no arbitrage possibilities.*

Identification of An Arbitrage Possibility

Proposition 1

Suppose that there exists a self-financed portfolio h, such that the value process V^h has the dynamics

$$dV^h(t) = k(t)V^h(t)dt,$$

where k is an adapted process. Then it must hold that $k(t) = r(t)$ for all t, or there exists an arbitrage possibility.

Schematic

We assume that the a priori given market consists of two assets with dynamics given by

$$\begin{aligned} dB(t) &= rB(t)dt, \\ dS(t) &= S(t)\alpha(t, S(t))dt + S(t)\sigma(t, S(t))d\bar{W}(t). \end{aligned} \tag{5}$$

We consider a simple contingent claim of the form

$$\mathcal{X} = \Phi(S(T)),$$

Its price process has the form

$$\Pi(t) = F(t, S(t)).$$

Schematic

Our problem is to find out what F must look like in order for the market $[S(t), B(t), \Pi(t)]$ to be free of arbitrage possibilities.

$$\begin{aligned} d\Pi(t) &= F_t dt + F_s dS(t) + \frac{1}{2} F_{ss} \sigma^2 S^2 dt \\ &= \frac{F_t + \alpha S F_s + \frac{1}{2} F_{ss} \sigma^2 S^2}{F} F dt + \frac{\sigma S F_s}{F} F d\bar{W}(t) \\ &= \alpha_\pi(t) \Pi(t) dt + \sigma_\pi(t) \Pi(t) d\bar{W}(t) \end{aligned} \tag{6}$$

Schematic

Consider a portfolio based on S and $\mathcal{X}$. Denote the relative portfolio by (u_s, u_π). By self-financing assumption,

$$dV^h(t) = V^h(t) \sum_{i=1}^{N} u_i(t) \frac{dS_i(t)}{S_i(t)}.$$

we have

$$dV = V\{u_s[\alpha dt + \sigma d\bar{W}] + u_\pi[\alpha_\pi dt + \sigma_\pi d\bar{W}]\}.$$

$$dV = V[u_s\alpha + u_\pi\alpha_\pi]dt + V[u_s\sigma + u_\pi\sigma_\pi]d\bar{W}].$$

Choose

$$\begin{aligned} u_s + u_\pi &= 1 \\ u_s\sigma + u_\pi\sigma_\pi &= 0 \end{aligned} \tag{7}$$

Then

$$dV = V[u_s\alpha + u_\pi\alpha_\pi]dt$$

Schematic

Use Proposition 1 to deduce that we must have the relation

$$u_s\alpha + u_\pi\alpha_\pi = r$$

From the linear system (7), we have the solution

$$\begin{aligned} u_s &= \frac{\sigma_\pi}{\sigma_\pi - \sigma} = \frac{SF_s}{SF_s - F} \\ u_\pi &= \frac{-\sigma}{\sigma_\pi - \sigma} = \frac{-F}{SF_s - F} \end{aligned} \tag{8}$$

Plugging these into $u_s\alpha + u_\pi\alpha_\pi = r$, we get that F has to satisfy the following PDE.

$$\begin{aligned} F_t(t,s) + rsF_s(t,s) + \frac{1}{2}s^2\sigma^2(t,s)F_{ss}(t,s) - rF(t,s) = 0 \\ F(T,s) = \Phi(s) \end{aligned} \tag{9}$$

Black-Scholes Equation

Theorem 1

The only pricing function which is consistent with the absence of arbitrage is when F is the solution of the following boundary value problem in the domain $[0, T] \times R_+$.

$$F_t(t,s) + rsF_s(t,s) + \frac{1}{2}s^2\sigma^2(t,s)F_{ss}(t,s) - rF(t,s) = 0$$
$$F(T,s) = \Phi(s) \tag{10}$$

A Corollary of Girsanov's Theorem

Let W be a standard BM on $(\Omega, \mathcal{F}, \mathcal{F}_t, \mathcal{P})$, and let $H \in \mathbb{L}$ be bounded. Let

$$X_t = \int_0^t H_s ds + W_t$$

and define Q by $\frac{dQ}{dP} = exp\{-\int_0^T H_s dW_s - \frac{1}{2}\int_0^T H_s^2 ds\}$, for some $T > 0$. Then, under Q, X is a SBM for $0 \leq t \leq T$.

Risk Neutral Valuation

Let us again consider a market given by the equations

$$\begin{aligned} dB(t) &= rB(t)dt, \\ dS(t) &= S(t)\alpha(t, S(t))dt + S(t)\sigma(t, S(t))d\bar{W}(t), \end{aligned} \tag{11}$$

Discounted price process $\tilde{S}(t) = e^{-rt}S(t)$. By Ito's formula,

$$\begin{aligned} d\tilde{S}(t) &= -re^{-rt}S(t)dt + e^{-rt}dS(t), \\ &= -r\tilde{S}(t)dt + \alpha\tilde{S}(t)dt + \sigma\tilde{S}(t)d\bar{(}W)(t) \\ &= (\alpha - r)\tilde{S}(t)dt + \sigma\tilde{S}(t)d\bar{(}W)(t) \end{aligned} \tag{12}$$

Risk Neutral Valuation

Under the mtg measure Q,

$$W_t^Q = \bar{W}_t + \frac{\alpha - r}{\sigma} t$$

is a SBM where

$$\frac{dQ}{dP} = exp\{-\int_0^T \frac{\alpha - r}{\sigma} dW_s - \frac{1}{2}\int_0^T (\frac{\alpha - r}{\sigma})^2 ds\}$$

Then,

$$d\tilde{S}(t) = \sigma \tilde{S}(t) dW^Q(t)$$

$$dS(t) = rS(t)dt + \sigma S(t) dW^Q(t)$$

Risk Neutral Valuation

Theorem 2 (Risk Neutral Valuation)

The arbitrage free price of the claim $\Phi(S(T))$ *is given by* $\Pi(t;\Phi) = F(t, S(t))$, *where* F *is given by the formula*

$$F(t,s) = e^{-r(T-t)} E^Q_{t,s}[\Phi(S(T))].$$

Proposition 2 (The Martingale Property)

In the Black–Scholes model, the price process $\Pi(t)$ *for every traded asset has the property that the normalized price process*

$$Z(t) = \frac{\Pi(t)}{B(t)}$$

is a martingale under the measure Q.

Derivation of the Pricing Formula

Recall

$$\begin{aligned} dS(t) &= rS(t)dt + \sigma S(t)dW^Q(t), \\ S(t) &= s \end{aligned} \tag{13}$$

We can write $S(T)$ as

$$S(T) = se^{(r-\frac{1}{2}\sigma^2)(T-t)+\sigma(W^Q(T)-W^Q(t))}$$

Thus we have the pricing formula

$$\begin{aligned} F(t,s) &= e^{-r(T-t)}E^Q[\Phi(se^{\sigma(W_T^Q-W_t^Q)+(r-\frac{1}{2}\sigma^2)(T-t)})], \\ &= e^{-r(T-t)}E^Q[\Phi(se^{\sigma\sqrt{T-t}Z+(r-\frac{1}{2}\sigma^2)(T-t)})], \end{aligned} \tag{14}$$

where $Z \sim N(0,1)$.

European Call

For an European call option, $\Phi(s) = (s - K)^+$. Define

$$d_1 = \frac{1}{\sigma\sqrt{T-t}}\left\{\ln\left(\frac{s}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)(T-t)\right\}, \tag{15}$$
$$d_2 = d_1 - \sigma\sqrt{T-t},$$

and $N(Z)$ is the cdf of $Z \sim N(0,1)$. Then

$$\begin{aligned} F(t,s) &= e^{-r(T-t)} E^Q[(se^{\sigma\sqrt{T-t}Z + (r-\frac{1}{2}\sigma^2)(T-t)} - K)^+] \\ &= e^{-r(T-t)} E^Q_{\{Z \geq -d_2\}}[(se^{\sigma\sqrt{T-t}Z + (r-\frac{1}{2}\sigma^2)(T-t)} - K)] \\ &= sE^Q_{\{Z \geq -d_2\}}[e^{\sigma\sqrt{T-t}Z - \frac{1}{2}\sigma^2(T-t)}] - Ke^{-r(T-t)}N(d_2) \end{aligned} \tag{16}$$

The Black-Sholes Formula

$$
\begin{aligned}
E^Q_{\{Z \geq -d_2\}}[e^{\sigma\sqrt{T-t}Z - \frac{1}{2}\sigma^2(T-t)}] &= \frac{1}{\sqrt{2\pi}} \int_{-d_2}^{\infty} e^{\sigma\sqrt{T-t}Z - \frac{1}{2}\sigma^2(T-t)} e^{-\frac{1}{2}z^2} dz \\
&= \frac{1}{\sqrt{2\pi}} \int_{-d_1}^{\infty} e^{-\frac{1}{2}u^2} du \\
&= N(d_1)
\end{aligned}
\tag{17}
$$

Thus

$$F(t,s) = sN(d_1) - e^{-r(T-t)}KN(d_2).$$

The above famous result is known as the **Black–Scholes formula**.

The Black-Sholes Formula

Proposition 3

The price of European call option with strike price K and time of maturity T is given by the formula $\Pi(t) = F(t, S(t))$, where

$$F(t,s) = sN[d_1(t,s)] - e^{-r(T-t)}KN[d_2(t,s)].$$

Here N is the cumulative distribution function for standard normal distribution and

$$\begin{aligned} d_1 &= \frac{1}{\sigma\sqrt{T-t}}\left\{\ln\left(\frac{s}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)(T-t)\right\}, \\ d_2 &= d_1 - \sigma\sqrt{T-t}. \end{aligned} \tag{18}$$

Volatility

In order to be able to use the theory derived above, we need to have numerical estimates of all the input parameters. s, r, T, t can be observed directly, which leaves us with the problem of obtaining an estimate of the volatility σ. Here there are two basic approaches, namely to use "historic volatility" or "implied volatility".

Historic Volatility

We observe the stock price process S at $t_0, t_1, \cdots, t_n$, where $\Delta t = t_i - t_{i-1}$. We thus observe $S(t_0), \cdots, S(t_n)$, and in order to estimate σ we use the fact that S has a log-normal distribution. Let us therefore define $\xi_1, \cdots, \xi_n$ by

$$\xi_i = \ln\left(\frac{S(t_i)}{S(t_{i-1})}\right).$$

We see that $\xi_1, \cdots, \xi_n$ are independent, normally distributed random variables with

$$E[\xi_i] = (\alpha - \frac{1}{2}\sigma^2)\Delta t,$$

$$Var[\xi_i] = \sigma^2 \Delta t.$$

Historic Volatility

Using elementary statistical theory we see that an estimate of σ is given by

$$\sigma^* = \frac{S_\xi}{\sqrt{\Delta t}}$$

where the sample variance S_ξ^2 is given by

$$S_\xi^2 = \frac{1}{n-1}\sum_{i=1}^{n}(\xi_i - \bar{\xi})^2,$$

$$\bar{\xi} = \frac{1}{n}\sum_{i=1}^{n}\xi_i.$$

Implied volatility

Suppose we want to value a European call with six months left to maturity. An argument against the use of historical volatility is that in real life volatility is not constant, but changes over time, and thus we should use the market expectation of the volatility for the next six months.
One way of finding the market expectation of the volatility is by getting market price data for another six month "benchmark" option, written on the same underlying stock as the option which we want to value. Denoting the price of the benchmark option by p, and writing the Black–Scholes pricing formula for European calls by $c(s, t, T, r, \sigma, K)$, we then solve the following equation for σ

$$p = c(s, t, T, r, \sigma, K).$$

This is called the implied volatility, and we then use the implied volatility for the benchmark in order to price our original option.

American Options

The options that can be exercised at any time prior to T is called an American contract. The holder will in fact choose a random exercise time τ and decide on an optimal exercise strategy.

$$\max_{\tau} E^{Q}[e^{-r\tau}\Phi(S_{\tau})].$$

Let us consider an American call option on a non-dividend paying underlying stock with final exercise date T and exercise price K. We denote the pricing function for the American option by $C(t,s)$ and the pricing function for the corresponding European option by $c(t,s)$.

American Options

Firstly,

$$C(t,s) \geq c(t,s).$$

Secondly,

$$c(t,s) \geq s - Ke^{-r(T-\tau)} > s - K, \ \forall t < T.$$

So we end up with the inequality

$$C(t,s) > s - K, \ \forall t < T.$$

The right-hand side gives us the value of actually exercising the option at time t. Since the value of the option is strictly greater than the value of exercising the option, it can thus not be optimal to exercise the option at any time $t < T$.

American Options

Proposition 4

Assume that $r > 0$. For an American call option, written on an underlying stock without dividends, the optimal exercise time τ is given by $\tau = T$. Thus the price of the American option coincides with the price of the corresponding European option.